

NLP Seminar Agentic Context Engineering

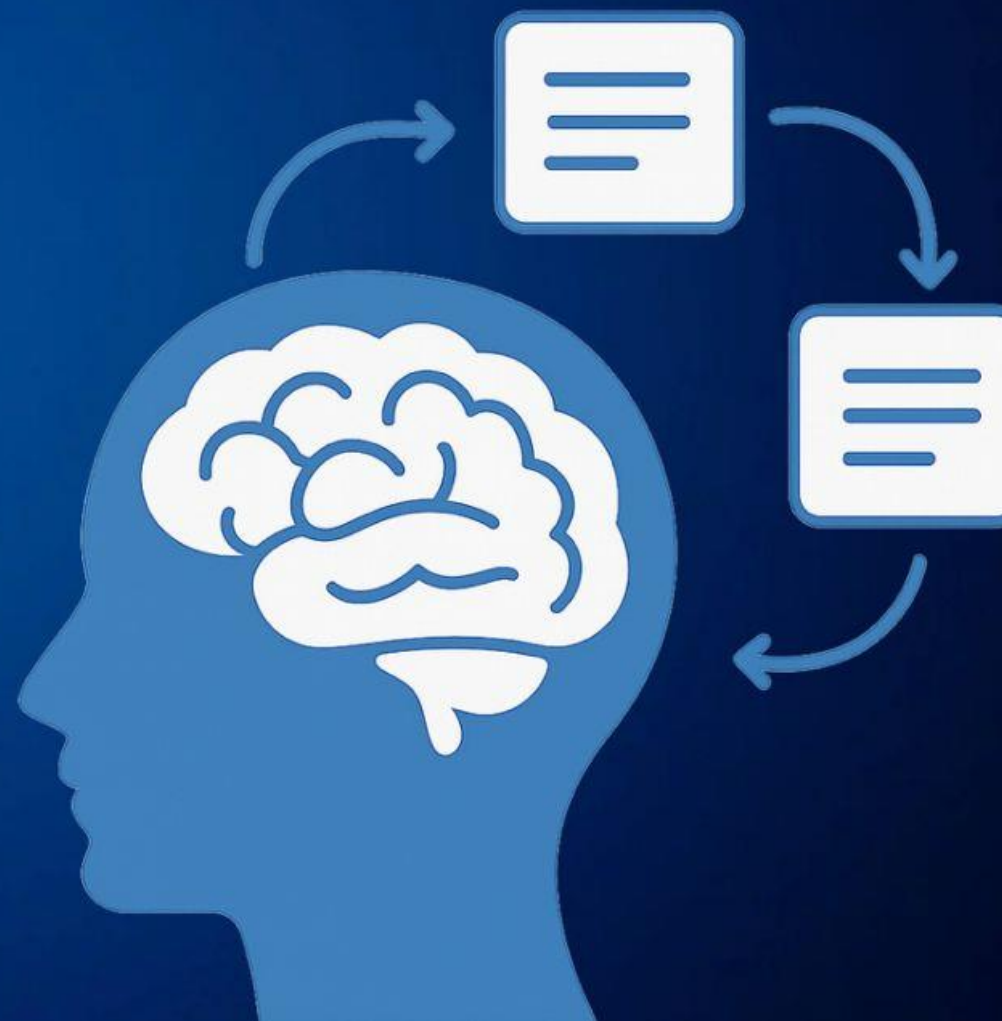
Evolving Contexts for Self-Improving Language Models

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Vamsidhar Kamanuru² Jay Rainton² Chen Wu² Mengmeng Ji² Hanchen Li³
Urmish Thakker² James Zou¹ Kunle Olukotun¹

¹ Stanford University ² SambaNova Systems, Inc. ³ UC Berkeley * equal contribution

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Presented By
Gil Caplan and Shachar Frenkel
Lecturer - Roi Reichart



Content



Introduction to LLM Agents



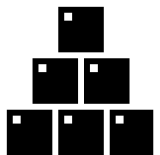
Presenting the Problem



Background in Literature



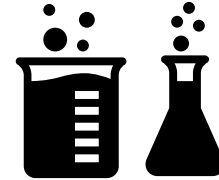
Research questions



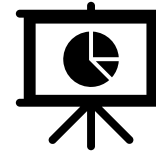
Main contributions



ACE model



Experiments



Results



Conclusions



Limitations + Ethical Issues

Content



Introduction to LLM Agents



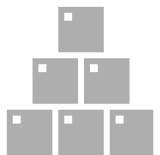
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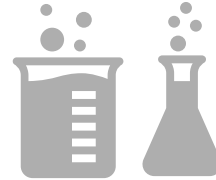
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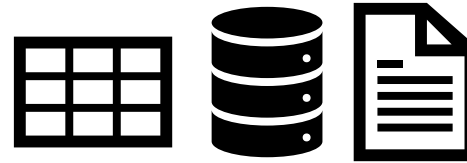
Conclusions



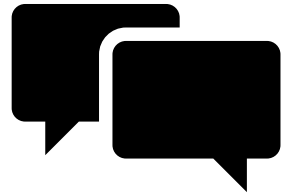
Limitations + Ethical Issues

Development of LLMs

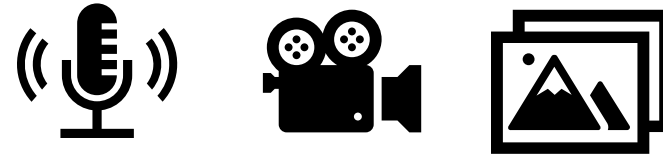
The NLP seminar
this semester is ____



Massive
amounts of data



Adapted to
conversational settings



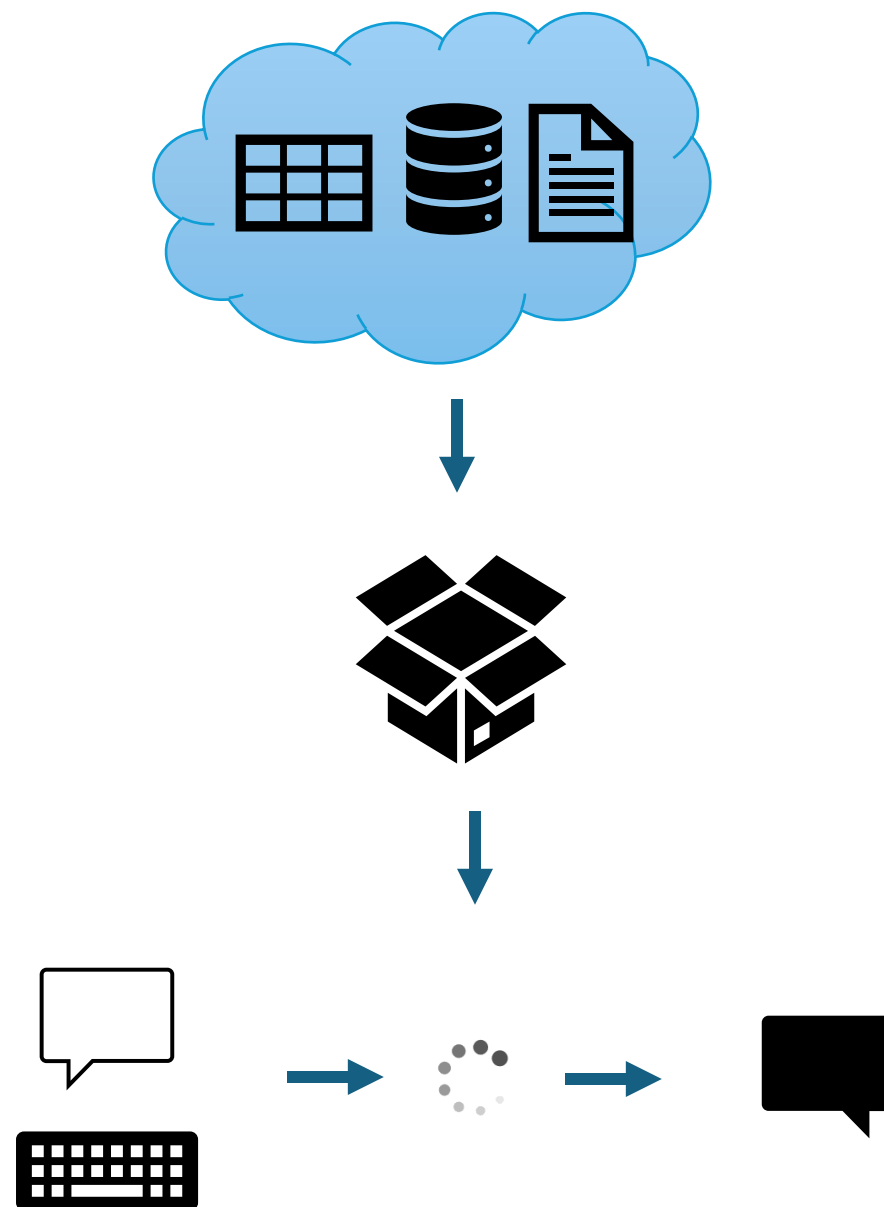
multimodeling



Reasoning
models

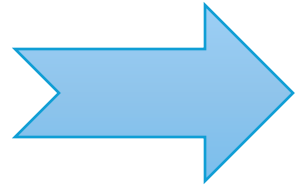
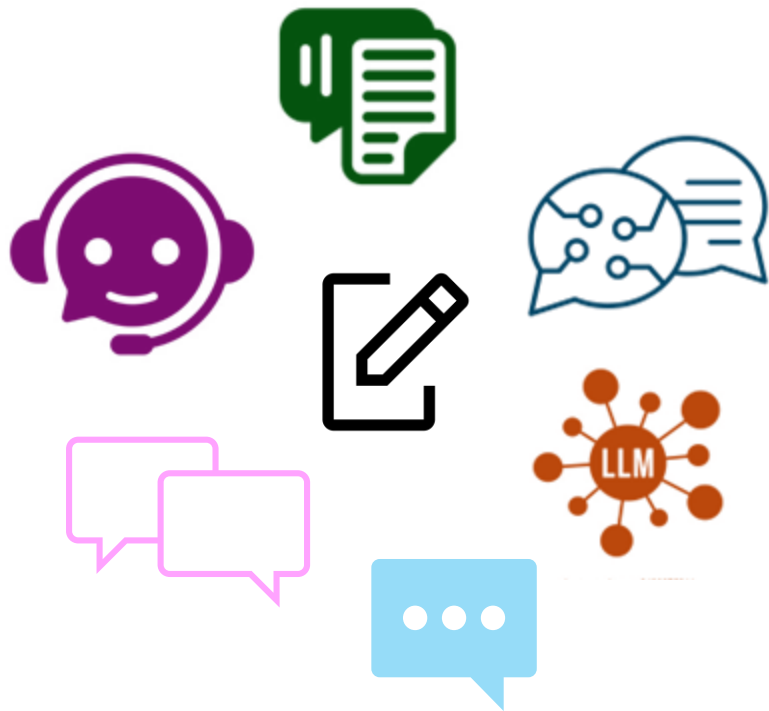
And still,
This is just...

And still, this is just...

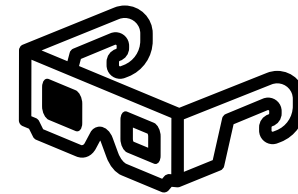
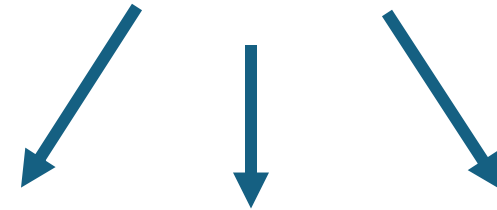


The next step - agents

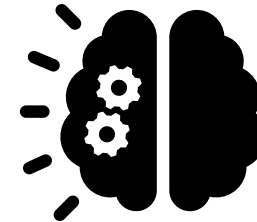
Agent - “agents are a system with complex reasoning capabilities, memory, and the means to execute tasks.”



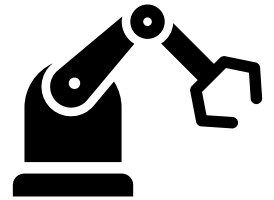
LLM Agent



Perceive the environment

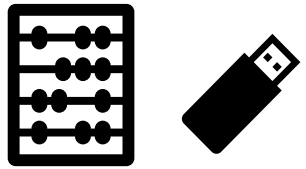


Reason



Autonomously
act in the
world

What do agents need?



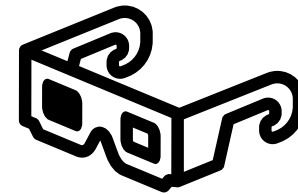
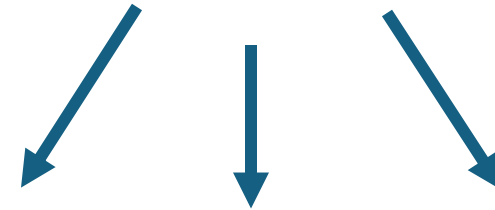
Memory
management



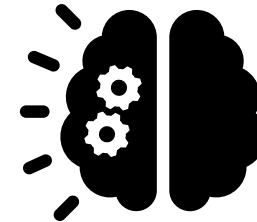
Prompts and
Instructions



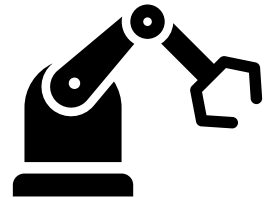
LLM Agent



Perceive the
environment

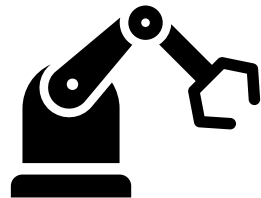


Reason



Autonomously
act in the
world

Need to be able to improve and maintain:



Content



Introduction to LLM Agents



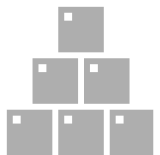
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Background in Literature



Research questions



Main contributions



ACE model



Experiments



Results

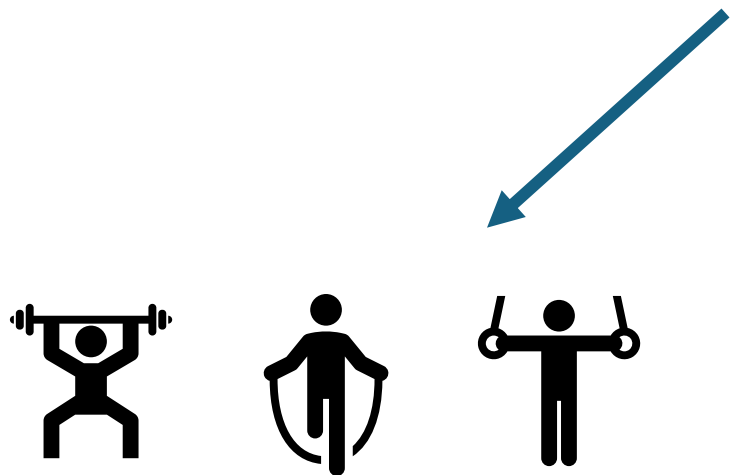


Conclusions



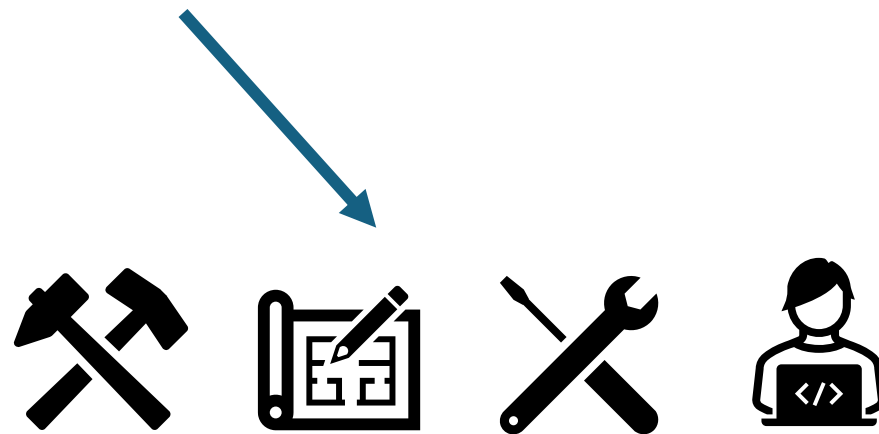
Limitations + Ethical Issues

Main purpose Is Improving LLM Agents



Training(weights), number of parameters, computational strength

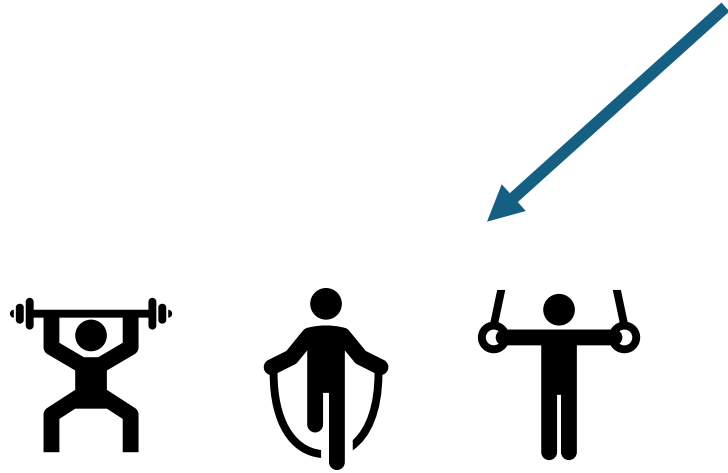
Traditional approach



Fixing/changing/improving/engineering the prompts

Context adaptation

Main purpose Is Improving LLM Agents



Training(weights), number of parameters, computational strength



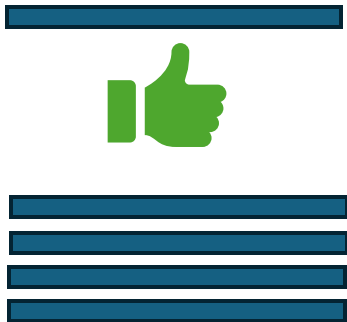
Fixing/changing/improving/
engineering the prompts

Context adaptation

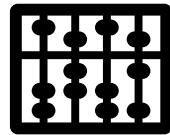
2 main problems

►► Brevity Bias

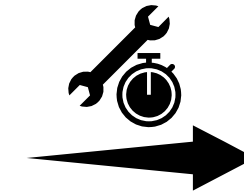
Brevity = shortness or conciseness of expression



Brevity bias leads to loss of important information



Context collapse

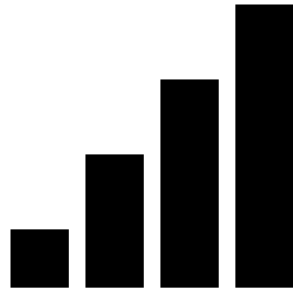
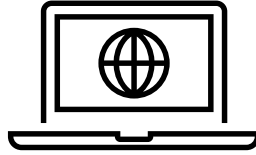


monolithic rewrite of the full context



Context collapse leading to sharp drop in performance

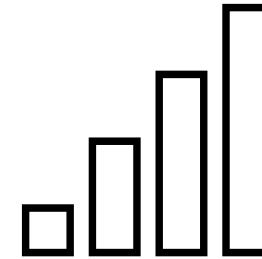
Offline vs Online



online

updating
context on
the fly

using
execution
feedback



offline

Training on a
closed set

Global fixed
system
prompt

Content



Introduction to LLM Agents



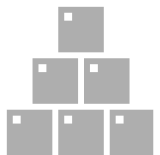
Presenting the Problem



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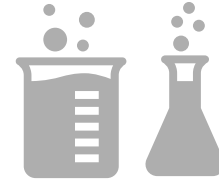
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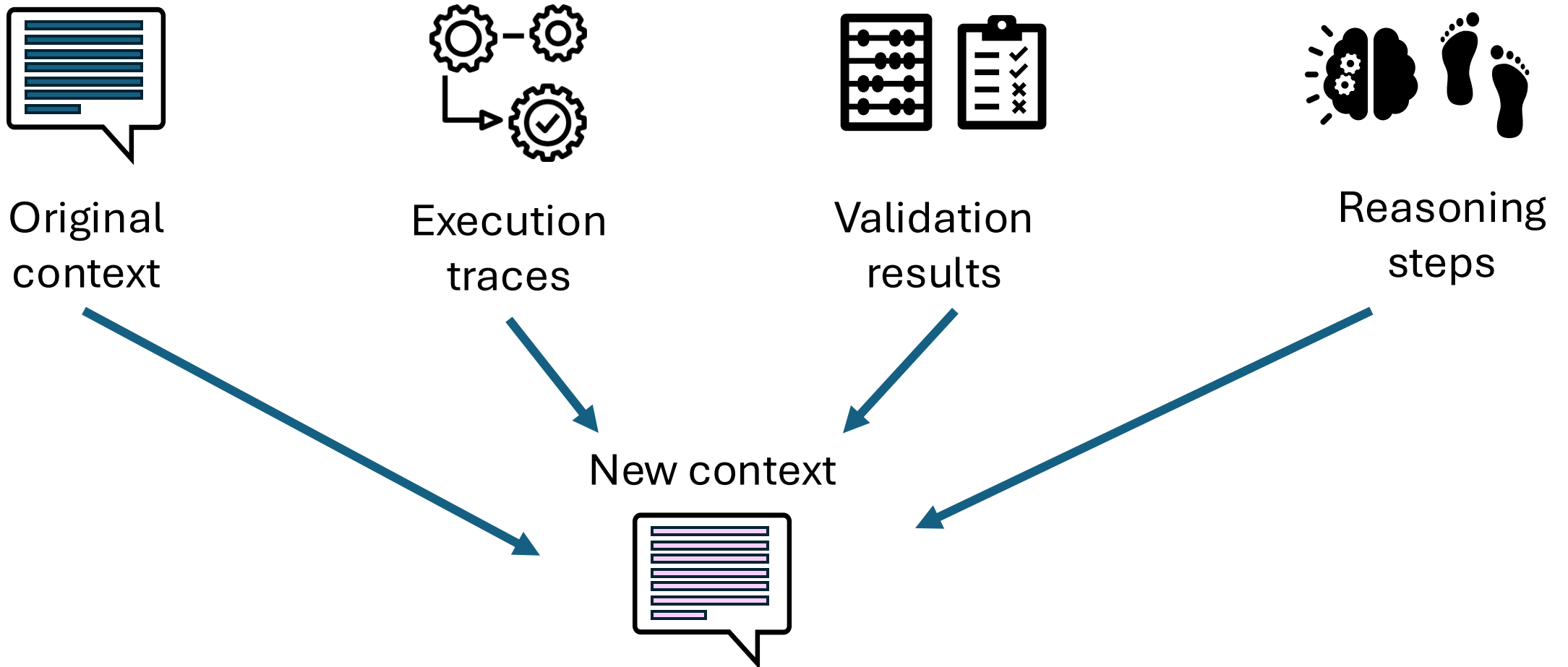
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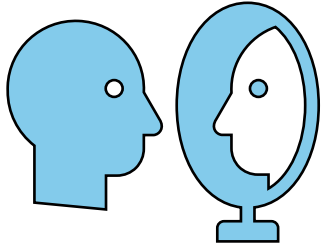
Limitations + Ethical Issues

Context Adaptation-

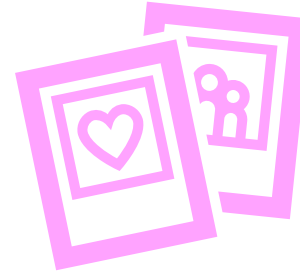
“modifying inputs with instructions, strategies or evidence, rather than weight updates.”



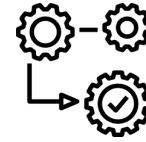
Methods



Reflection



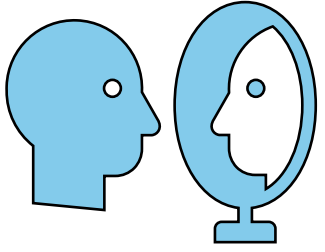
GEPA



TextGrad

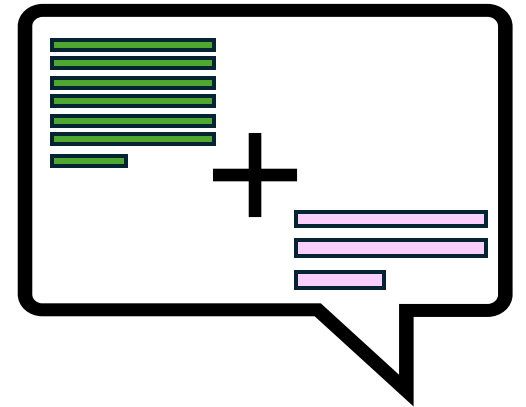
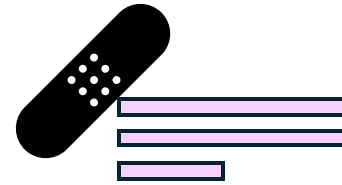
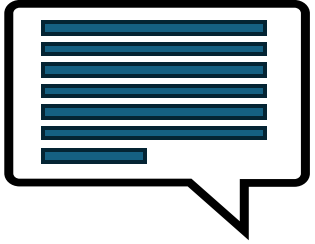


Dynamic
Cheatsheetc



Reflection

context



New task context

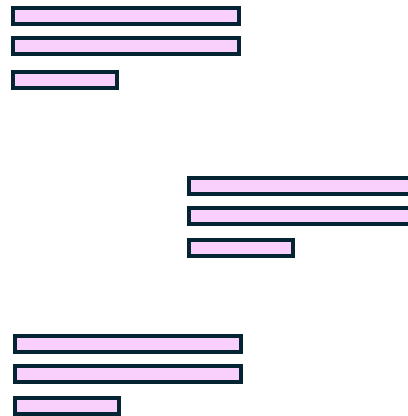
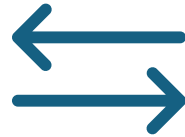
New task



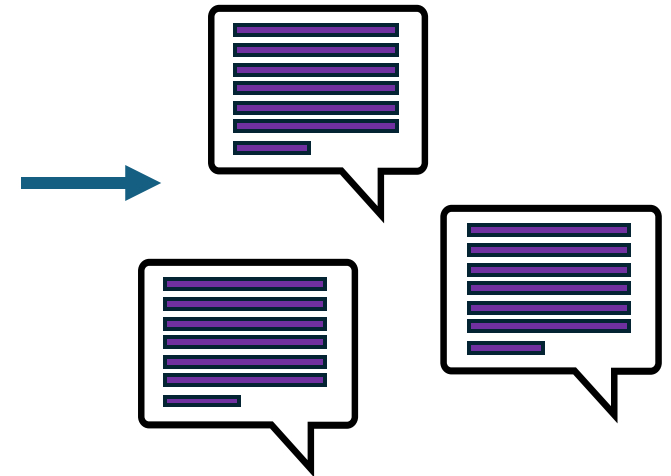
GEPA



Population of
candidate prompts



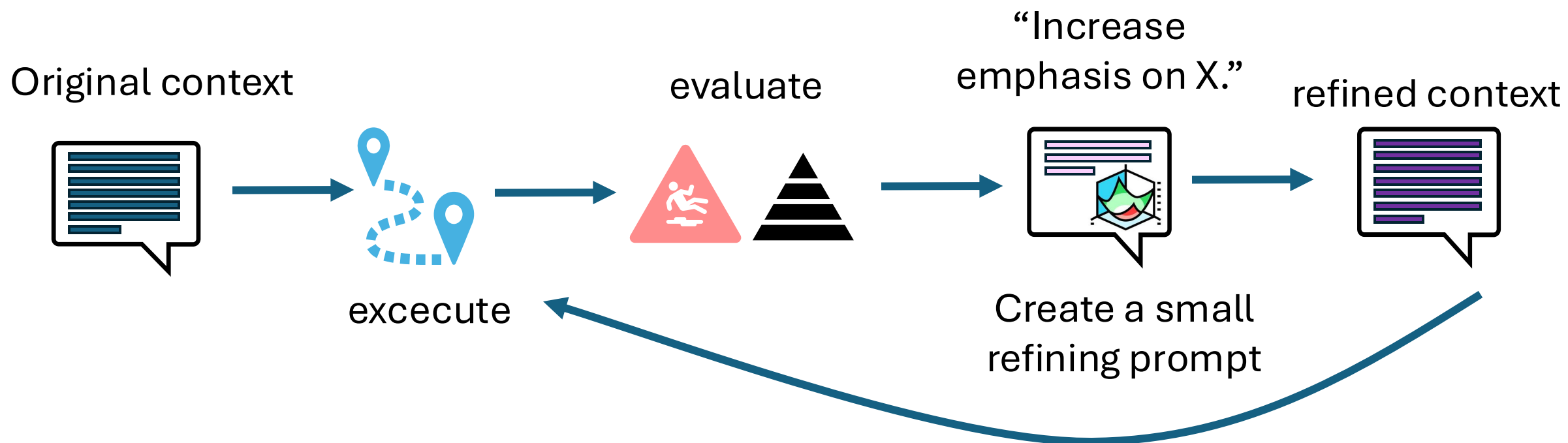
Insights and
suggestions –
What went wrong?



Best refined prompts
stay



TextGrad

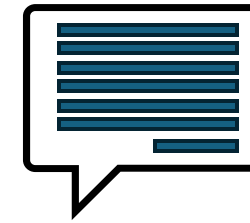




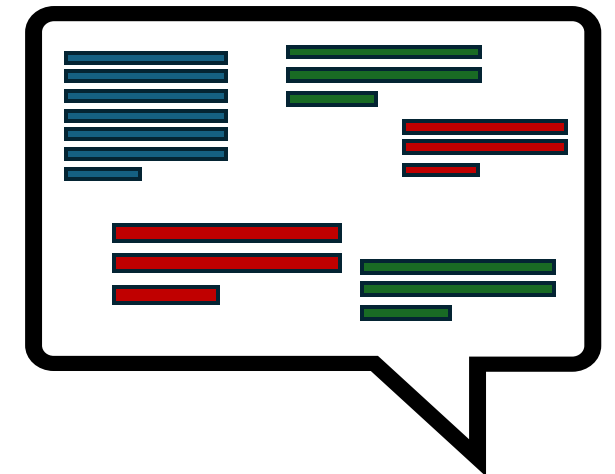
Dynamic Cheetsheet



New task



Retrieving the
most relevant
paragraphs

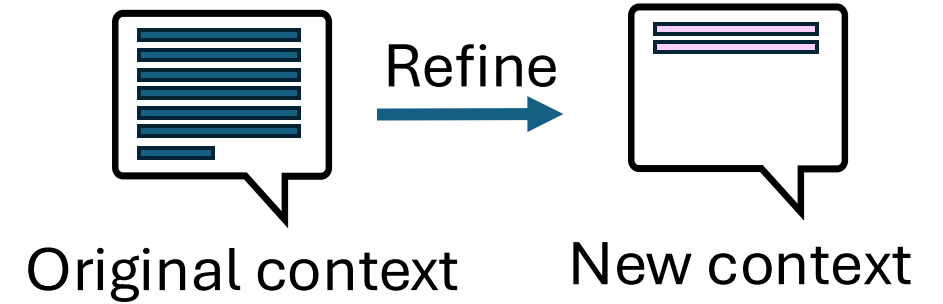


Current literature limitations



Brevity Bias –

“The tendency of optimization to collapse toward short, generic prompts



~~DIVERSITY~~

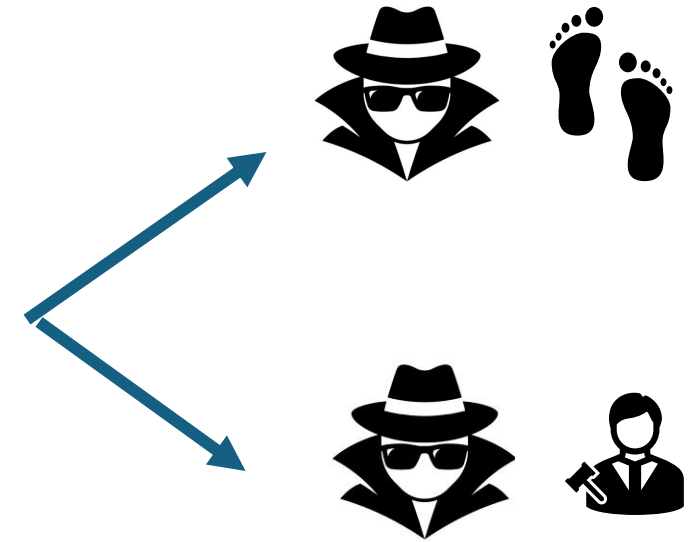
Losing diversity



Losing domain-specific detail



Reoccurring errors



Performance declines for

Current literature limitations



Context collapse

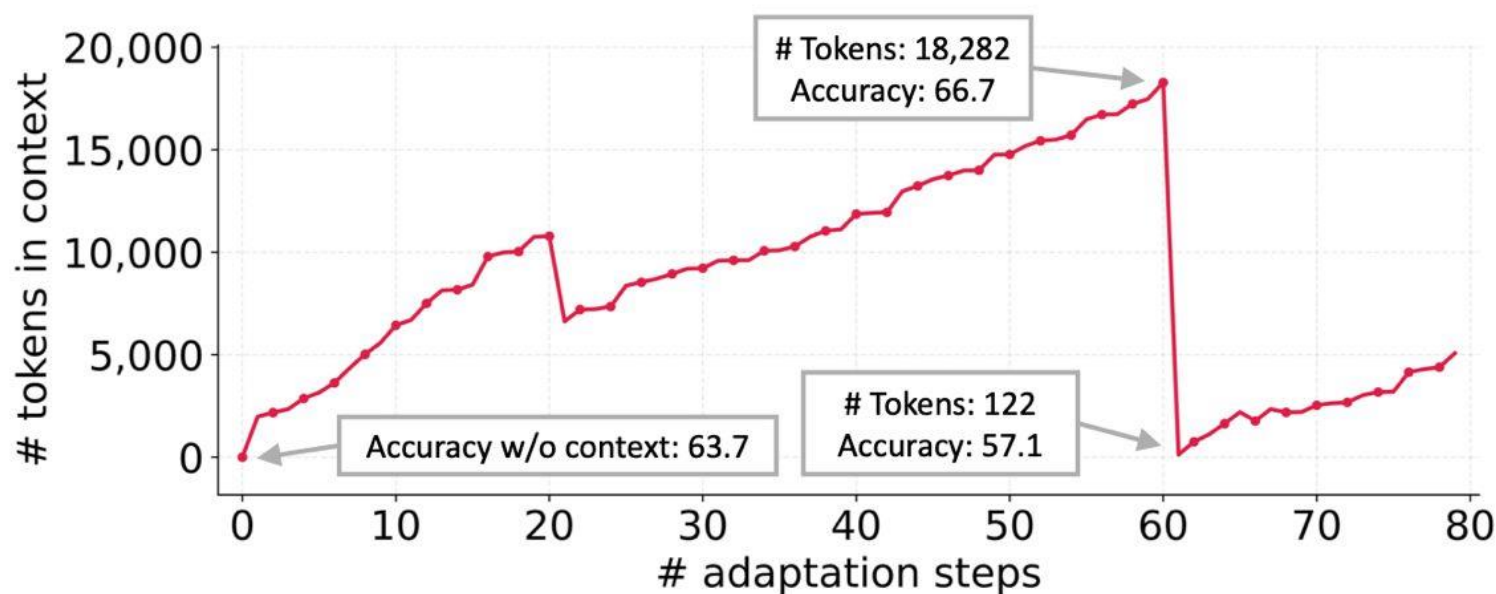
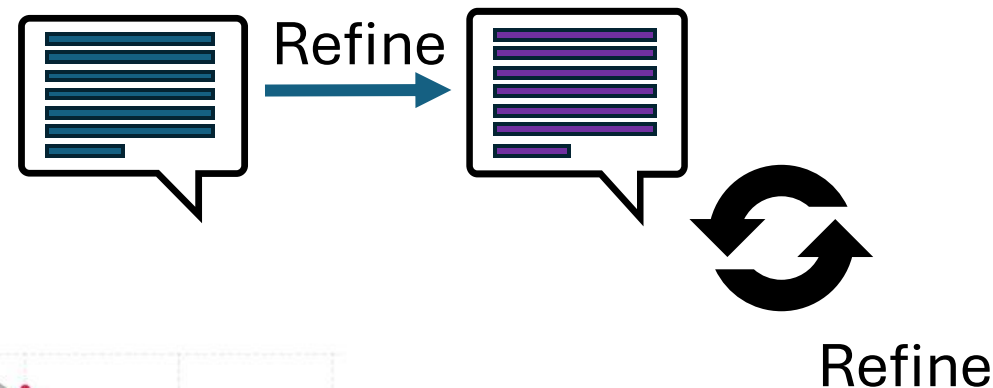


Figure 2: **Context Collapse.** Monolithic rewriting of context by an LLM can collapse it into shorter, less informative summaries, leading to sharp performance drops.

“We argue that contexts should function not as concise summaries, but as comprehensive, evolving playbooks—detailed, inclusive, and rich with domain insights.”



Content



Introduction to LLM Agents



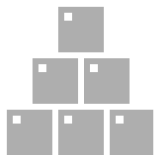
Presenting the Problem



Background in Literature



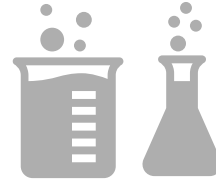
Research questions



Main contributions



ACE model



Experiments



Results



Conclusions



Limitations+ Ethical Issues

Research questions

- Can an LLM agent improve itself over time solely by evolving its context?
- What is an effective representation of an agent's context?
- How does context granularity affect agent performance?
- How can the context-building process be organized such that it reliably accumulates domain-specific knowledge?
- Does using execution-based signals improve agent performance, without ground truth labels?
- Which ACE components drive most of the performance gains?



Content



Introduction to LLM Agents



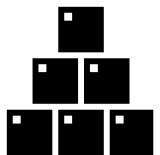
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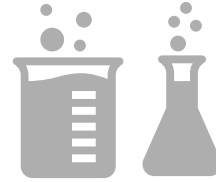
Research questions



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Research Main Contributions

- Can an LLM agent improve itself over time solely by evolving its context?

Yes – the writers present an agentic framework for context adaptations

- What is an effective representation of an agent's context?

Contribution: Assigning separate Models tasks and integrating them into as a framework to represent the tasks context accurately

- How does context granularity affect agent performance, without ground truth labels?

** Brevity bias may still exist in each insight itself

too much specify can lead to overfitting, whilst generalization to underfitting to given task

Research Main Contributions

- How can the context-building process be organized such that it reliably accumulates domain-specific knowledge?

The results will indicate Ace roles improve on tasks that require accumulation of knowledge

- Does using execution-based signals improve agent performance, without ground truth labels?

Apparently, they don't isolate to cases based on execution-based signals

Main Contributions - Spoiler

ACE optimizes contexts both offline (e.g., system prompts) and online (e.g., agent memory), consistently outperforming strong baselines: +10.6% on agents and +8.6% on finance based tasks!

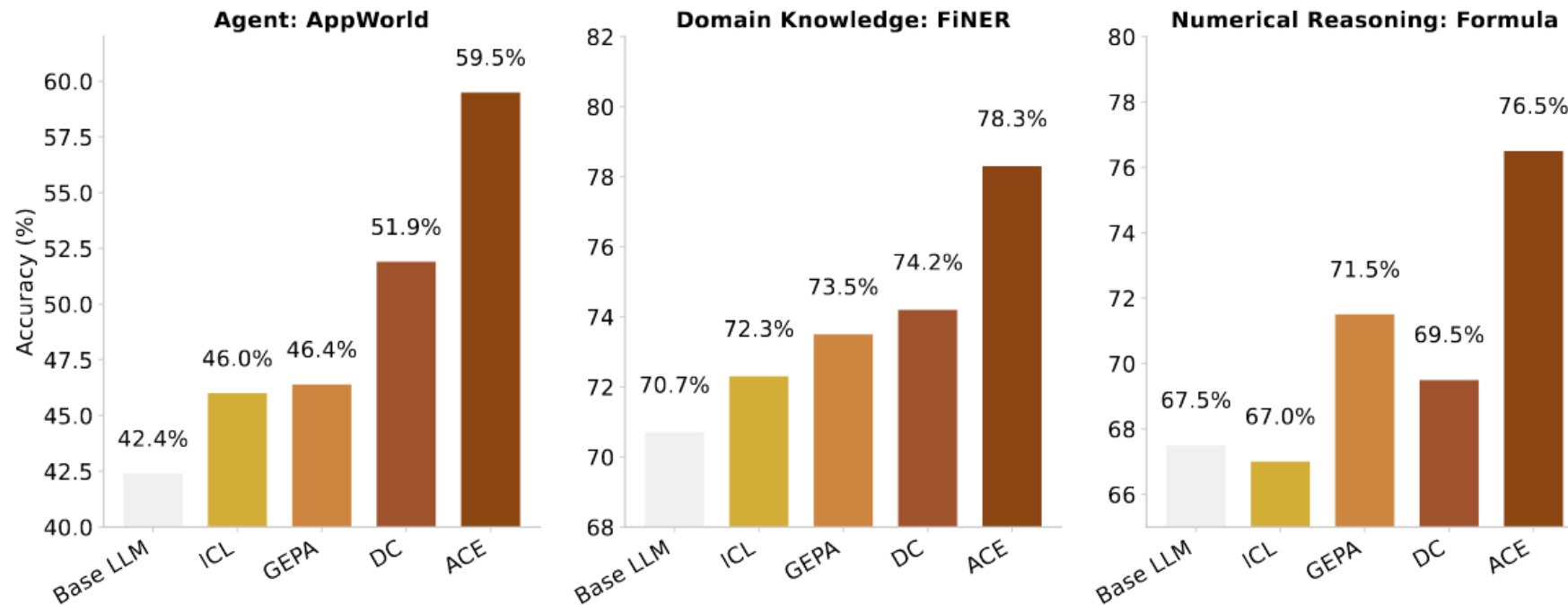


Figure 1: **Overall Performance Results.** Our proposed framework, ACE, consistently outperforms strong baselines across agent and domain-specific reasoning tasks.

Agentic Context Engineering (ACE)

Method Contributions



Structured knowledge - Context is broken into discrete, tracked items



Delta contexts- small sets of candidate text integrated by ACE framework

Content



Introduction to LLM Agents



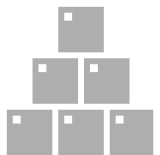
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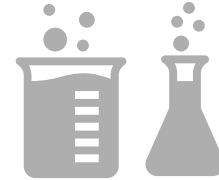
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Limitations + Ethical Issues

Agentic Context Engineering (ACE) - Agent roles



The generator



The reflector



The curator

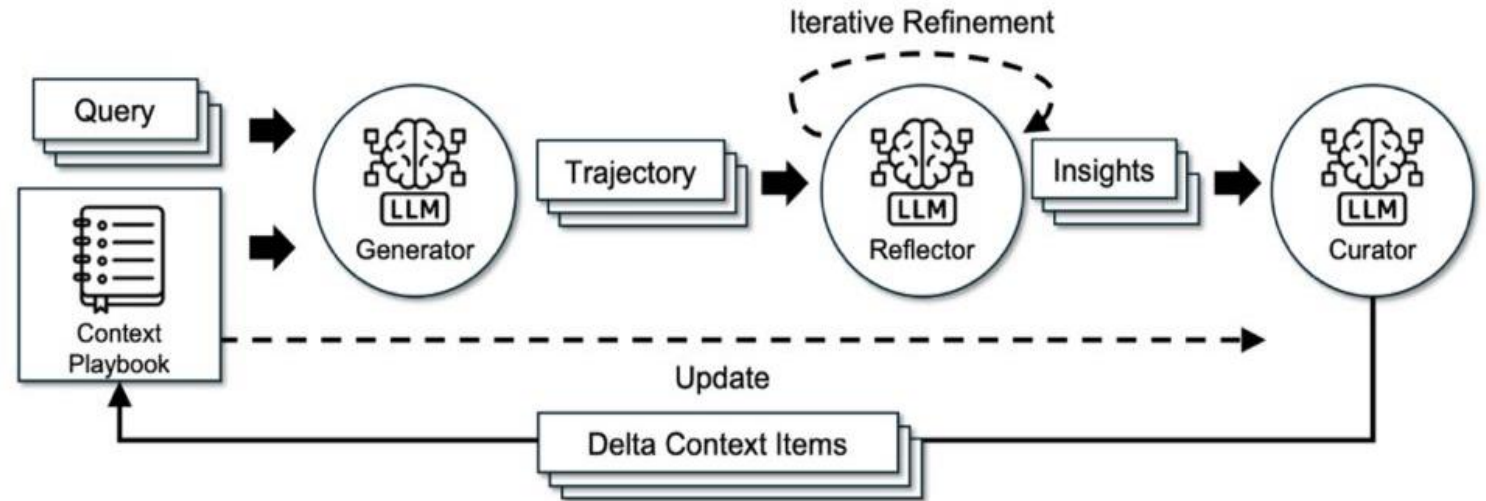


Figure 4: **The ACE Framework.** Inspired by Dynamic Cheatsheet, ACE adopts an agentic architecture with three specialized components: a Generator, a Reflector, and a Curator.



The Generator

**Remember
Cheetsheet?

producing reasoning trajectories for new queries, which
surface both effective strategies and recurring pitfalls.

Trajectories

Effective
strategies



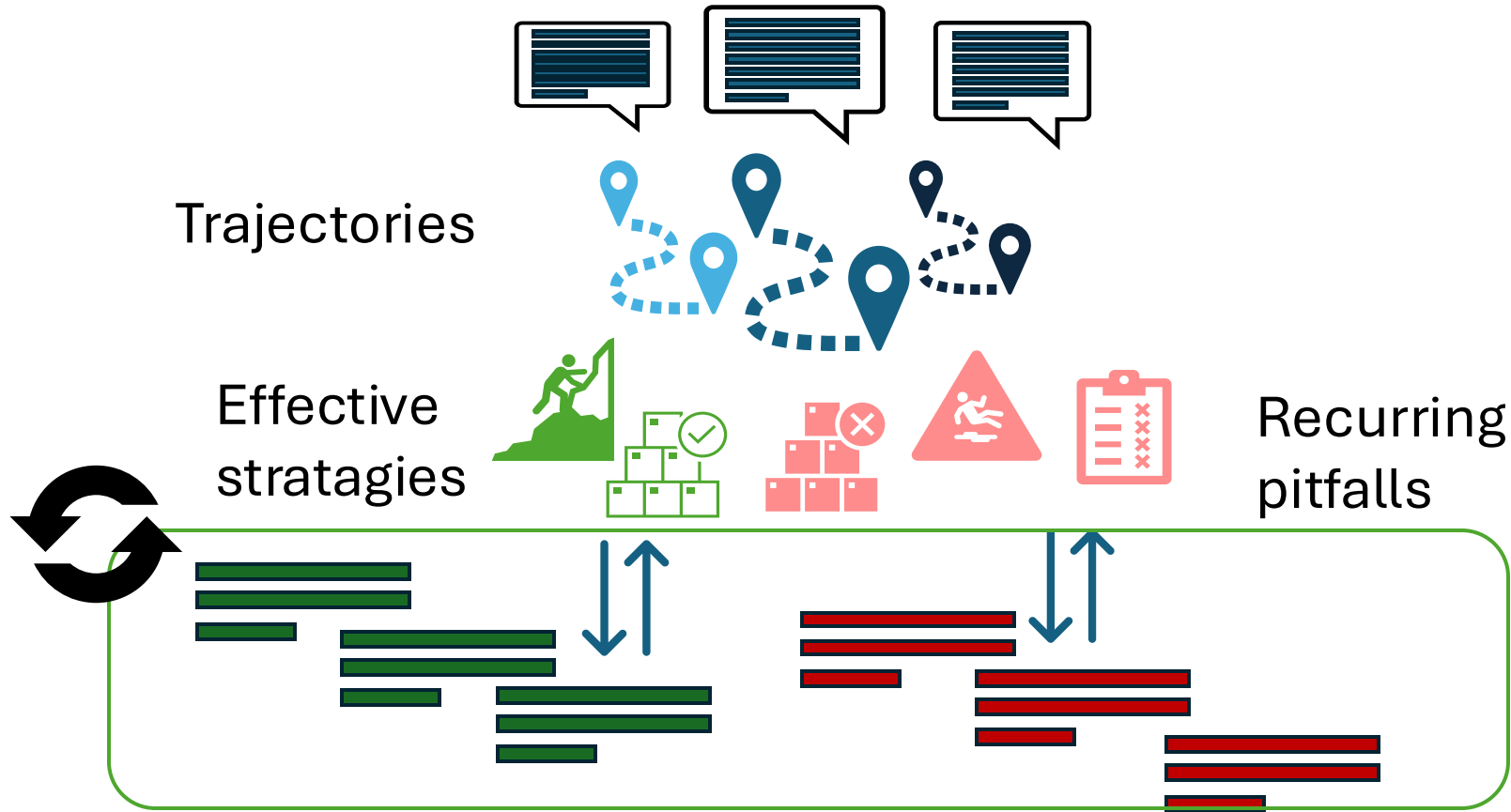
Recurring
pitfalls



The Reflector

**Remember
Cheetsheet?

critiques these traces to extract lessons, optionally
refining them across multiple iterations

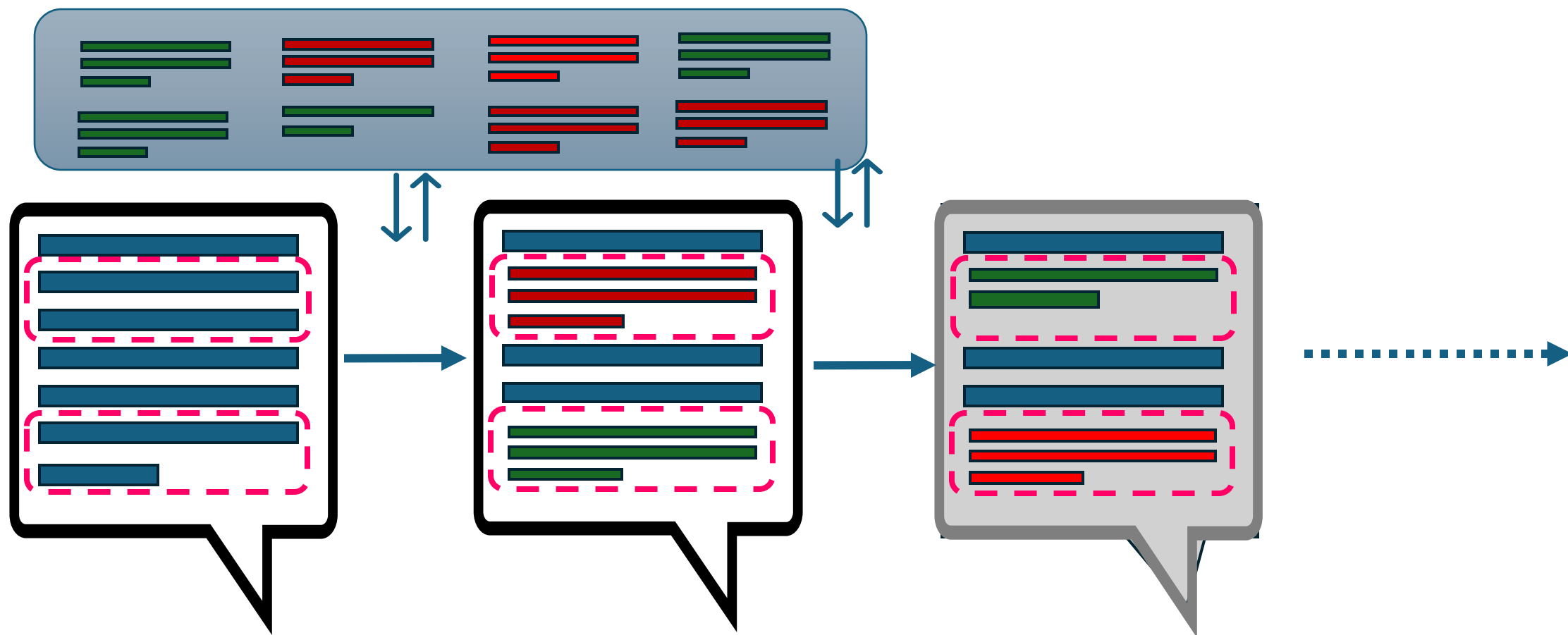




The Curator

**Remember
Cheat sheet?

The Curator then synthesizes these lessons into compact delta entries, which are merged deterministically into the existing context



Key innovations



Dedicated reflector - adding a separate step before curation that extracts independent insights

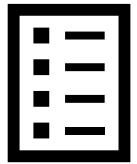


Incremental delta updates - replace costly monolithic rewrites with localized edits, reducing both latency and compute cost

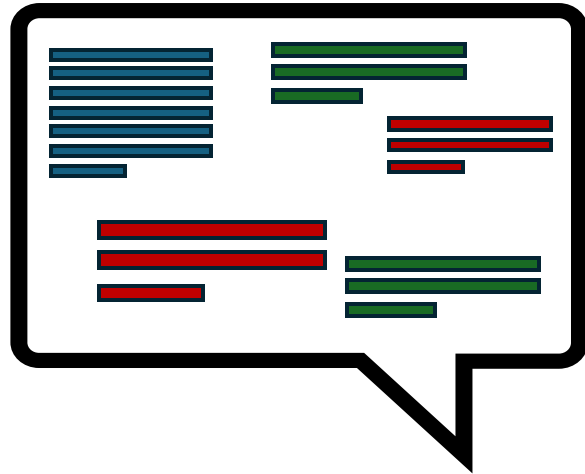


Grow and Refine mechanism - balances steady context expansion with redundancy control.

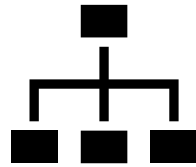
Incremental delta updates



Bullet points



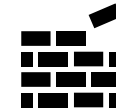
**Memory entries
(Dynamic Cheatsheet)**



Representation:

1. Metadata

2. Content



Benefits:

1. Localization

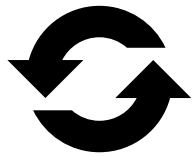
2. Fine-grained retrieval

3. Incremental adaptation

Grow and Refine mechanism



Appending Itemized bullets with new ids

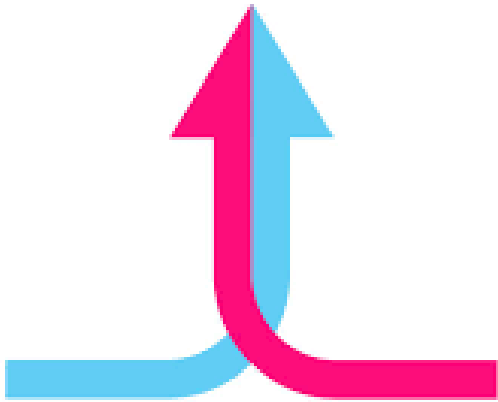


Updating existing itemized bullets by modifying in place

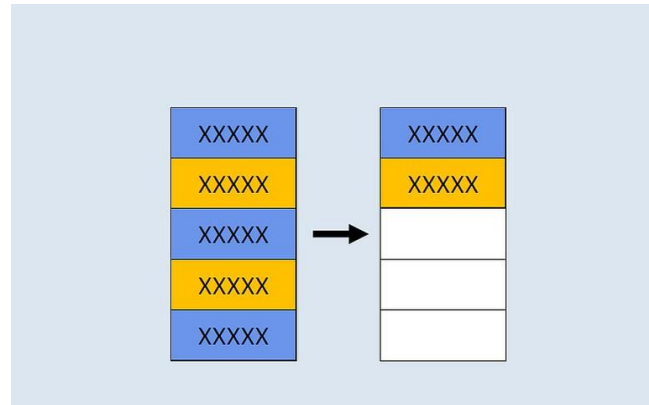


Prune redundant bullets

Incremental adaptation



Merging



De-duplication



pruning

Content



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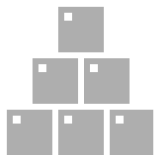
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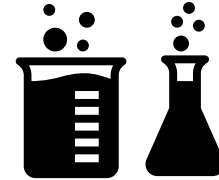
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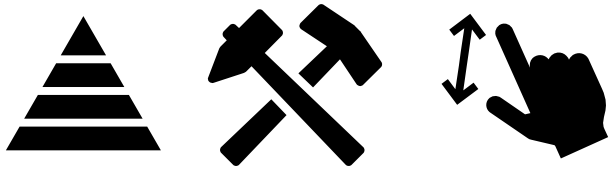
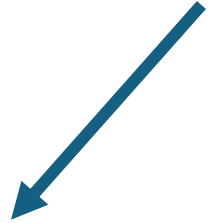


Conclusions



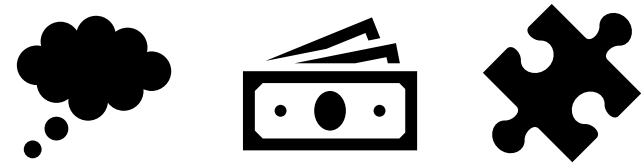
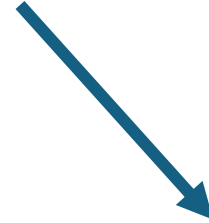
Limitations + Ethical Issues

Experiments - Tasks



Agent benchmark

General tasks - AppWorld



Domain-specific benchmarks

Financial Analysis:

FiNeR + Formula

Experiments - AppWorld



750 tasks across everyday scenarios



Evaluation is measured by TGC/SGC

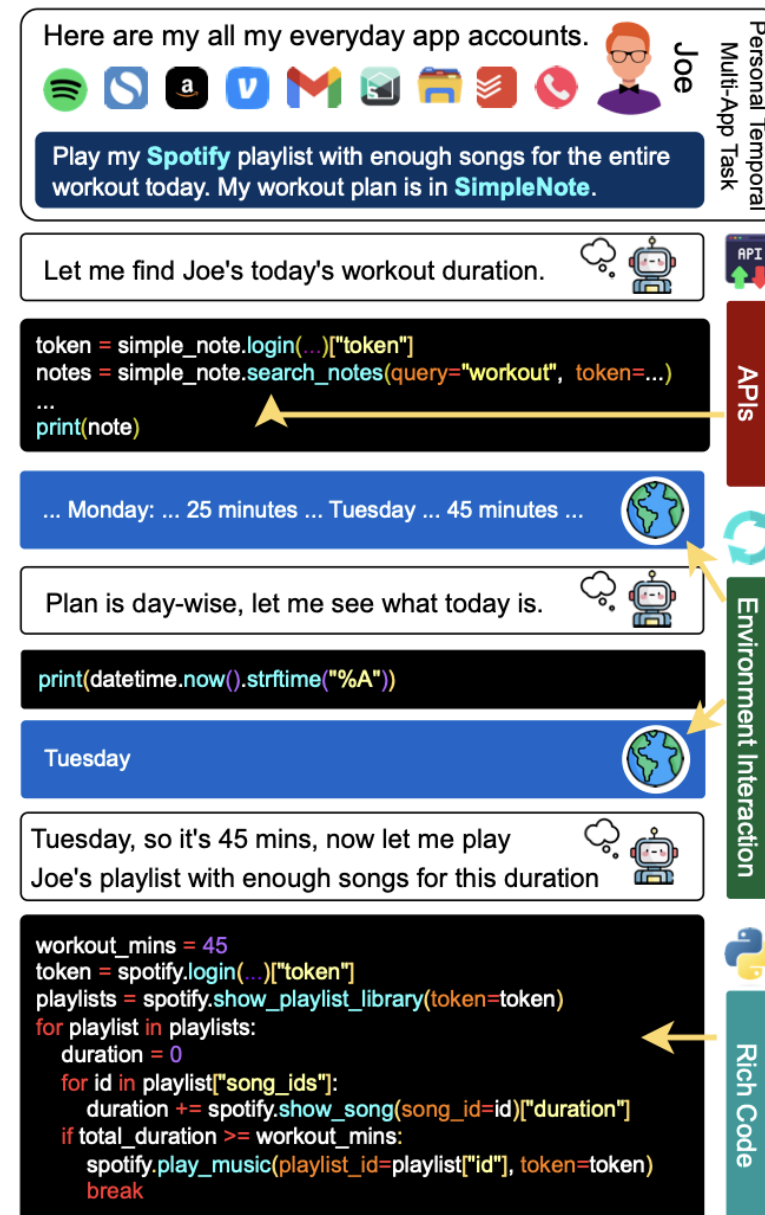


Figure 1: A (shortened) day-to-day task from AppWorld Benchmark requiring **rich code** with **environment interaction** and **API calls**.²

Experiments – FiNeR/Formula

FiNeR- Task is XBRL tagging for financial reports

Formula- Task is to extract financial information and computing in a financial formula

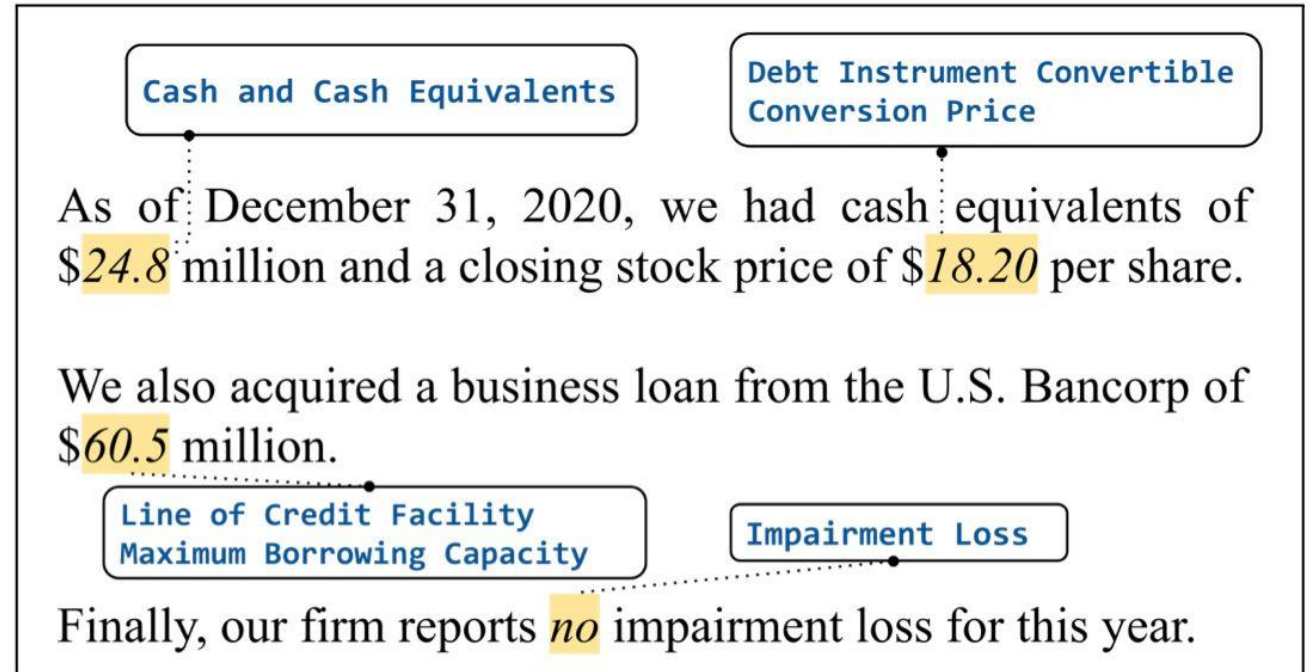


Figure 1: Sentences from FiNER-139, with XBRL tags on numeric and non-numeric tokens. XBRL tags are actually XML-based and most tagged tokens are numeric.

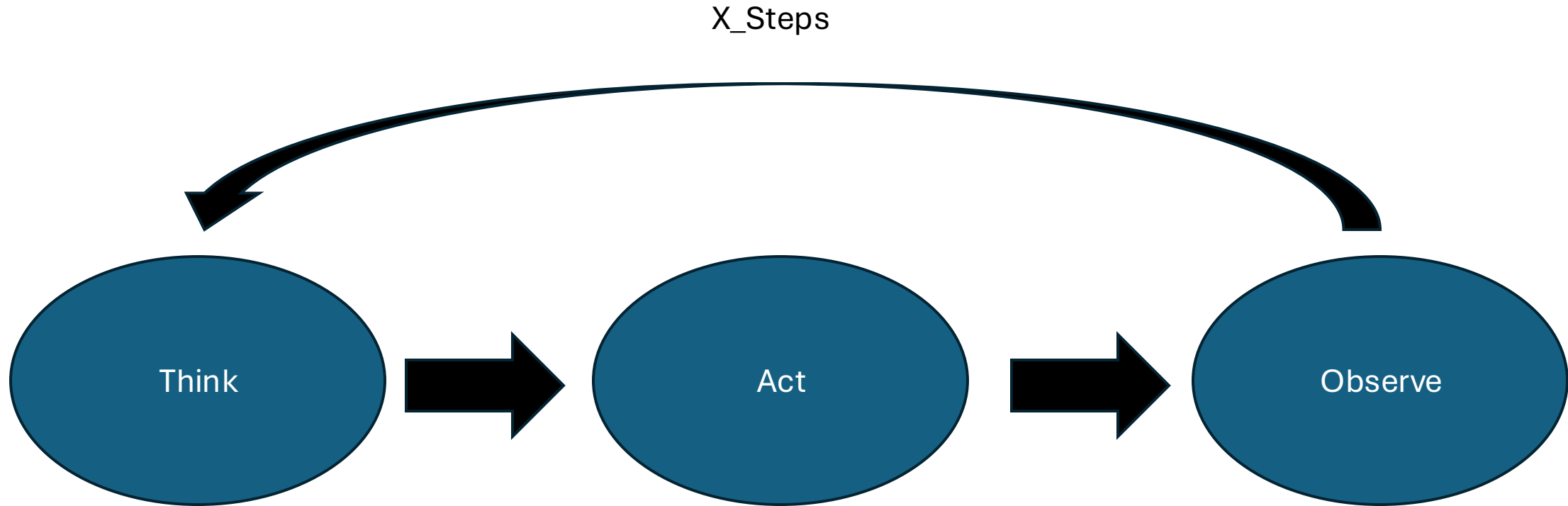
Experiments - Baselines

- **Baseline LLM**
- **ICL (in context learning)**
- **MICROv2**

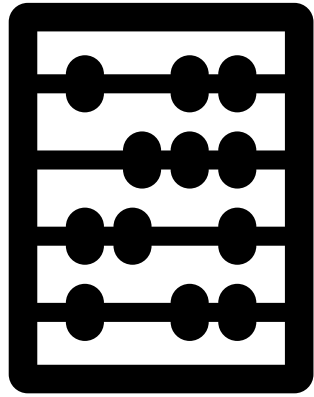
Remember-

- **GEPA**
- **DC (Dynamic Cheatsheet)**
- **ACE (paper's method)**

ReAct – Reason and Act Agent framework

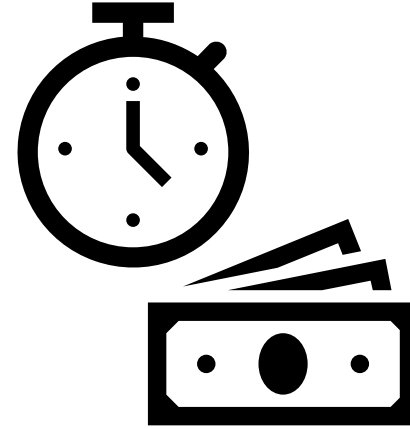


More experiments



Ablation studies – **AppWorld**

Individual design choices of ACE framework
contribute to context adaptation on task



Cost and speed analysis

Measuring the context adaptation latency,
number of rollouts, and dollar costs of ACE

Content



Introduction to LLM Agents



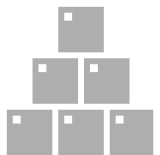
Presenting the Problem



Background in Literature



Research questions



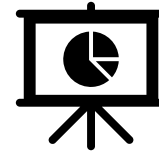
Main contributions



ACE model



Experiments



Results



Conclusions



Limitations + Ethical Issues

Results - AppWorld

Method	GT Labels	Test-Normal		Test-Challenge		Average
		TGC↑	SGC↑	TGC↑	SGC↑	
DeepSeek-V3.1 as Base LLM						
ReAct		63.7	42.9	41.5	21.6	42.4
Offline Adaptation						
ReAct + ICL	✓	64.3 _{+0.6}	46.4 _{+3.5}	46.0 _{+4.5}	27.3 _{+5.7}	46.0 _{+3.6}
ReAct + GEPA	✓	64.9 _{+1.2}	44.6 _{+1.7}	46.0 _{+4.5}	30.2 _{+8.6}	46.4 _{+4.0}
ReAct + ACE	✓	76.2 _{+12.5}	64.3 _{+21.4}	57.3 _{+15.8}	39.6 _{+18.0}	59.4 _{+17.0}
ReAct + ACE	✗	75.0 _{+11.3}	64.3 _{+21.4}	54.4 _{+12.9}	35.2 _{+13.6}	57.2 _{+14.8}
Online Adaptation						
ReAct + DC (CU)	✗	65.5 _{+1.8}	58.9 _{+16.0}	52.3 _{+10.8}	30.8 _{+9.2}	51.9 _{+9.5}
ReAct + ACE	✗	69.6 _{+5.9}	53.6 _{+10.7}	66.0 _{+24.5}	48.9 _{+27.3}	59.5 _{+17.1}

Table 1: **Results on the AppWorld Agent Benchmark.** "GT labels" indicates whether ground-truth labels are available to the Reflector during adaptation. We evaluate the ACE framework against multiple baselines on top of the official ReAct implementation, both for offline and online context adaptation. ReAct + ACE outperforms selected baselines by an average of 10.6%, and could achieve good performance even without access to GT labels.

Results - FiNeR

Method	GT Labels	FINER (Acc↑)	Formula (Acc↑)	Average
DeepSeek-V3.1 as Base LLM				
Base LLM		70.7	67.5	69.1
Offline Adaptation				
ICL	✓	72.3 _{+1.6}	67.0 _{-0.5}	69.6 _{+0.5}
MIPROv2	✓	72.4 _{+1.7}	69.5 _{+2.0}	70.9 _{+1.8}
GEPA	✓	73.5 _{+2.8}	71.5 _{+4.0}	72.5 _{+3.4}
ACE	✓	78.3_{+7.6}	85.5_{+18.0}	81.9_{+12.8}
ACE	✗	71.1 _{+0.4}	83.0 _{+15.5}	77.1 _{+8.0}
Online Adaptation				
DC (CU)	✓	74.2 _{+3.5}	69.5 _{+2.0}	71.8 _{+2.7}
DC (CU)	✗	68.3 _{-2.4}	62.5 _{-5.0}	65.4 _{-3.7}
ACE	✓	76.7_{+6.0}	76.5 _{+9.0}	76.6_{+7.5}
ACE	✗	67.3 _{-3.4}	78.5_{+11.0}	72.9 _{+3.8}

Table 2: **Results on Financial Analysis Benchmark.** "GT labels" indicates whether ground-truth labels are available to the Reflector during adaptation. With GT labels, ACE outperforms selected baselines by an average of 8.6%, highlighting the advantage of structured and evolving contexts for domain-specific reasoning. However, we also observe that in the absence of reliable feedback signals (*e.g.*, ground-truth labels or execution outcomes), both ACE and other adaptive methods such as Dynamic Cheatsheet may degrade, suggesting that context adaptation depends critically on feedback quality.

Ablation studies on AppWorld

Method	GT Labels	Test-Normal		Test-Challenge		Average
		TGC↑	SGC↑	TGC↑	SGC↑	
DeepSeek-V3.1 as Base LLM						
ReAct		63.7	42.9	41.5	21.6	42.4
Offline Adaptation						
ReAct + ACE w/o Reflector or multi-epoch	✓	70.8 _{+7.1}	55.4 _{+12.5}	55.9 _{+14.4}	38.1 _{+17.5}	55.1 _{+12.7}
ReAct + ACE w/o multi-epoch	✓	72.0 _{+8.3}	60.7 _{+17.8}	54.9 _{+13.4}	39.6 _{+18.0}	56.8 _{+14.4}
ReAct + ACE	✓	76.2 _{+12.5}	64.3 _{+21.4}	57.3 _{+15.8}	39.6 _{+18.0}	59.4 _{+17.0}
Online Adaptation						
ReAct + ACE	✗	67.9 _{+4.2}	51.8 _{+8.9}	61.4 _{+19.9}	43.2 _{+21.6}	56.1 _{+13.7}
ReAct + ACE + offline warmup	✗	69.6 _{+5.9}	53.6 _{+10.7}	66.0 _{+24.5}	48.9 _{+27.3}	59.5 _{+17.1}

Table 3: **Ablation Studies on AppWorld.** We study how particular design choices of ACE (iterative refinement, multi-epoch adaptation, and offline warmup) could help high-quality context adaptation.

Cost and Speed Analysis

Method	Latency (s)↓	# Rollouts↓
ReAct + GEPA	53898	1434
ReAct + ACE	9517 _(-82.3%)	357 _(-75.1%)

(a) **Offline** (AppWorld).

Method	Latency (s)↓	Token Cost (\$)↓
DC (CU)	65104	17.7
ACE	5503 _(-91.5%)	2.9 _(-83.6%)

(b) **Online** (FiNER).

Table 4: **Cost and Speed Analysis.** We measure the context adaptation latency, number of rollouts, and dollar costs of ACE against GEPA (offline) and DC (online).

Content



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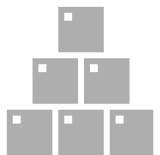
Presenting the Problem



Background in Literature



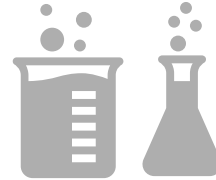
Research questions



Main contributions



ACE model



Experiments



Results



Conclusions



Limitations + Ethical Issues

Conclutions

- ACE enables agents to self-improve by dynamically refining their input context.

boosts accuracy on the **AppWorld benchmark** by up to **17.1%**

- Large Gains on Domain-Specific Benchmarks.

On complex **financial reasoning benchmarks**, ACE delivers an average performance gain of **8.6%** over strong baselines

- Effective by Design

According to **ablation studies**, **each component increases performance**

- Lower Cost and Adaptation Latency.

reducing **adaptation latency** by **86.9%** on average,

Contributions

- ACE consistently outperforms strong baselines,

ACE is an effective representation of an agent's context

- ACE is able to construct effective contexts without labeled supervision, instead leveraging execution feedback and environment signals

key ingredients for self-improving LLMs and agents.

- Longer Context != Higher Serving Cost

Was able to use a longer context without exploding cost

Content



Introduction to LLM Agents



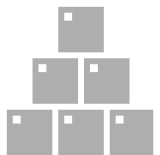
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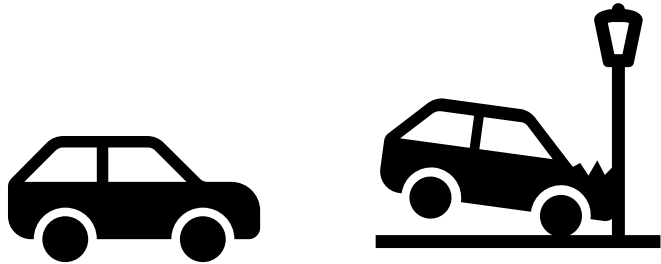
Conclusions



Limitations + Ethical Issues

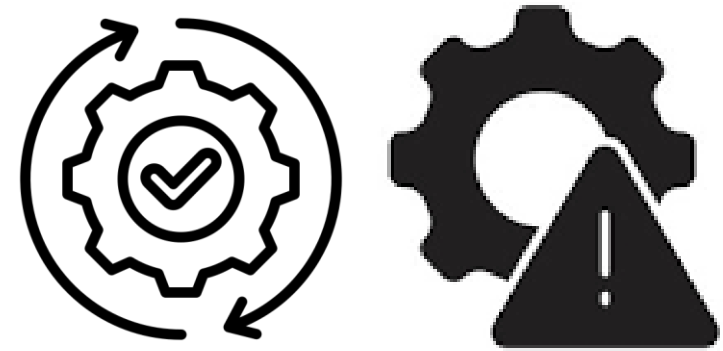
Limitations

1



Strong dependency on
the **Reflector's** quality

2



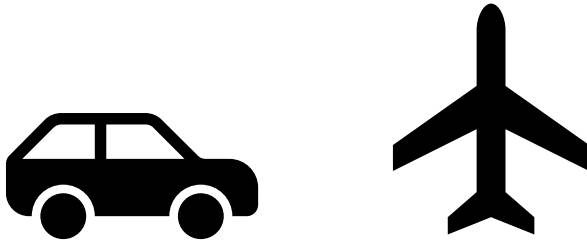
Strong dependency on
on good **feedback**

Needs strong,
meaningful
execution signals

Sensitive to
missing or
wrong signals

Limitations

3

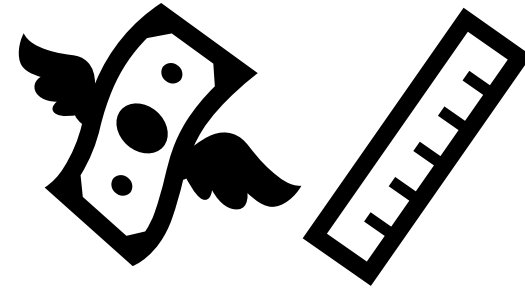


A _____ B

Unfit for simpler tasks

** Brevity bias is
sometimes better

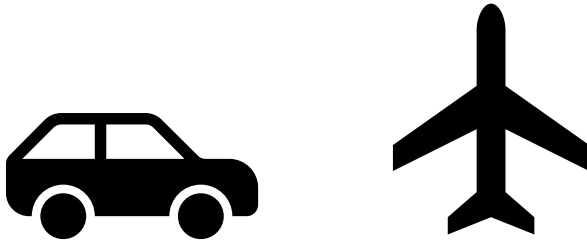
4



Cost and scalability concerns

Limitations

3

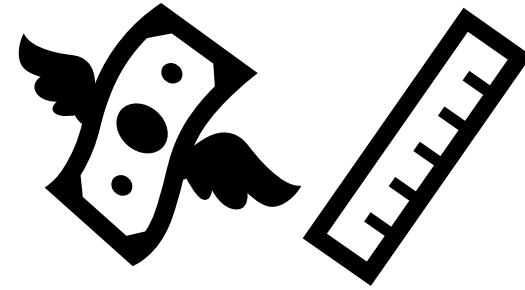


A _____ B

Unfit for simpler tasks

** Brevity bias is
sometimes better

4



Cost and scalability concerns

The paper claims:

Longer Context != Higher Serving Cost.

Reality:



Ethical Issues

- Stereotype Biases
- Privacy
- Transparency and Misinformation





You did good Ace